

Three ways to search binary embeddings

From stored data to query latency,
memory usage, and recall

- A** Cluster, then score every selected document.
- B** Cluster, then traverse an inverted bitplane index.
- C** Cluster, then enumerate keys from the query's ideal sign pattern.

This report develops the three methods from the same score and the same retrieval task. It explains what is built before a query, what is touched during a query, and where each method can lose recall. Every general cost formula uses named parameters. Small numerical examples make the formulas concrete; measured results are identified separately.

Reading order. Start with the four-bit example and shared pipeline. Each method then has its own data layout, pseudocode, operation count, latency model, and memory calculation. The statistical section explains what can and cannot be inferred about recall. The final section collects the formulas for comparison under a RAM limit.

Contents

1	Start with the question a search must answer	2
2	What is built once, and what every query pays	6
3	Approach A: cluster, then full scan	10
4	Approach B: cluster, then inverted bitplanes and branching	14
5	Approach C: cluster, then walk backwards from the ideal key	18
6	Deriving recall without assuming the answer	26
7	What the existing measurements support	31
8	Bring the formulas together	34
A	A finite-sample view of prefix recall	39
B	Checklist for an honest numerical implementation	39
C	Floating-point arithmetic and exactness	40
D	Optional expansion of the embedding cost	40
E	Reproduction and parameter ledger	41

1 Start with the question a search must answer

Suppose a database contains N documents. A query arrives, and the system must return K documents. We want the smallest query latency that fits the available RAM and reaches a specified recall target. Before comparing algorithms, we must agree on what “the correct K documents” means.

This report first uses one precise target: the top K documents under the same **float-query, binary-document inner product**. This isolates the search index. A later reranking step may use longer float embeddings, but that is a different score and a different reference ranking. Exact binary search need not recover the exact float ranking, and neither ranking is automatically the same as human relevance.

1.1 Three costs that must not be confused

1. **Index storage:** data built in advance and kept between queries. Examples are packed document codes, cluster offsets, centroids, bitplanes, and key directories.
2. **Query RAM:** temporary live data needed by one query. Examples are a query vector, score tables, a top- R result heap, pending branch masks, and key-enumeration states.
3. **Query work:** operations and data accesses over time. Reading the same array a thousand times does not require a thousand stored copies, but it can cost a thousand reads.

The number of scored documents alone does not measure work. A method can score fewer documents and still read more memory or manage a much larger queue.

1.2 The shared pipeline

All three primary methods begin with clusters. The no-clustering experiment is the special case $C = P = 1$, not a different definition of the task.

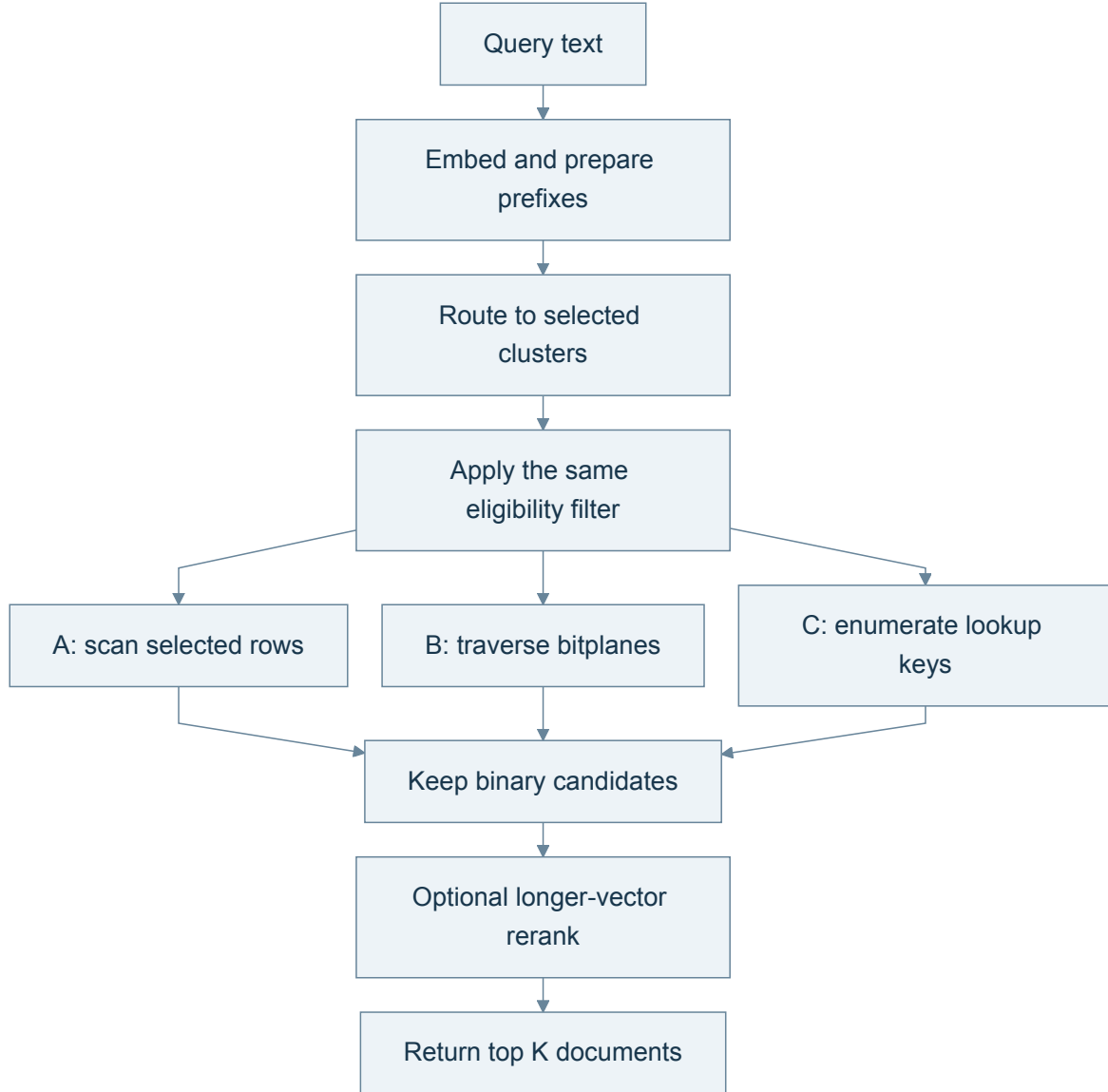


Figure 1: The common query pipeline. Only the candidate-search stage changes between A, B and C.

An embedding model converts query text to a float vector. A short representation routes the query to selected clusters. Optional filters select eligible documents. The index produces up to R binary-score candidates, with $R \geq K$. An optional final stage fetches longer vectors and reranks those candidates, then returns K results.

If the experiment only measures the index, cache the query embedding and omit the optional reranker. If the experiment claims end-to-end query latency, include them. We will keep both versions visible in the formulas.

1.3 A four-bit example we can calculate by hand

Let

$$q = (0.7, -0.4, 0.2, -0.1), \quad t = \text{sign}(q) = (+1, -1, +1, -1).$$

Write a stored $+1$ as bit 1 and -1 as bit 0. The ideal bit pattern is therefore 1010. Define $a_i = |q_i|$ and $A = \sum_i a_i = 1.4$. A document bit that agrees with the query contributes $+a_i$; a disagreement contributes $-a_i$.

$$S(q, b) = \sum_{i=1}^d q_i b_i = A - 2 \underbrace{\sum_{i=1}^d a_i \mathbf{1}[b_i \neq t_i]}_{\pi_q(b)}. \quad (1)$$

The quantity $\pi_q(b)$ is the **weighted mismatch penalty**. Maximizing score is exactly the same as minimizing this penalty. The factor two is not a hardware setting: replacing a contribution $+a_i$ by $-a_i$ changes the score by $2a_i$.

Document	Code	Mismatching coordinates	Score
1	1010	none	1.4
2	1011	4	1.2
3	1000	3	1.0
4	0010	1	0.0
5	1110	2	0.6
6	0101	1, 2, 3, 4	-1.4
7	1001	3, 4	0.8
8	0011	1, 4	-0.2

For example, document 7 pays $0.2 + 0.1 = 0.3$, so its score is $1.4 - 2(0.3) = 0.8$. The exact global top two are documents 1 and 2. We break equal scores by a stable numeric document ID throughout. This is the internal ID used by the index; an external identifier or URL can be recovered through metadata.

The query gives both a preferred sign and a cost for getting that sign wrong. It does *not* tell us which preferred patterns exist in the database. That distinction drives both branching and key enumeration.

1.4 Matryoshka prefixes are useful; they are not sorted magnitudes

A Matryoshka model trains selected prefixes of an embedding to remain useful representations [2]. It does not impose $|q_1| \geq |q_2| \geq \dots \geq |q_D|$ for every query. A later coordinate may have a larger magnitude than an earlier one. Nor does the name of the training method tell us the numerical recall of the first h binary signs.

There are three distinct choices:

- Take the first h coordinates: a fixed prefix that can be indexed in advance.
- Sort all coordinates by $|q_i|$: a query-dependent traversal order, useful for bitplanes.
- Use a longer representation for final scoring: this can change the ranking after retrieval.

A hash table built for one fixed prefix does not become a table for the query’s largest-magnitude coordinates just because the query is sorted. The lookup coordinates would have changed.

1.5 Parameters and units

The formulas below are not tied to a particular code length, cluster count, word width, or SIMD implementation. Table 1 is the common notation. Additional symbols are introduced where their algorithm first needs them.

Table 1: Shared parameters. Counts and byte sizes are different types of quantity.

Symbol	Meaning	Units
N	Number of stored documents	documents
D	Full embedding dimension	coordinates
d	Dimension of the binary score in Equation (1)	bits / doc
d_r, d_f	Routing and optional final-reranking dimensions	coordinates
C	Number of disjoint clusters	clusters
n_j	Physical number of document rows in cluster j ; $\sum_j n_j = N$	documents
$\mathcal{P}(q), P$	Probed cluster set and its size	set, count
e_j	Eligible rows in probed cluster j , after filters; $e_j \leq n_j$	documents
$N_{\text{sel}}, N_{\text{elig}}$	$\sum_{j \in \mathcal{P}} n_j$ and $\sum_{j \in \mathcal{P}} e_j$	documents
K, R	Final result count and binary candidate-heap capacity, $K \leq R$	documents
w, b_w	Bits and bytes per bitmap word; $b_w = w/8$	bits, bytes
g, G	Bits per score-table group and $G = \lceil d/g \rceil$ groups	bits, groups
b_s, b_q	Bytes per stored table score and query float	bytes
b_{id}, b_o	Bytes per document ID and offset	bytes
b_c, b_f	Bytes per routing-centroid and final-vector coordinate	bytes
M_{RAM}	Available physical RAM budget	bytes
J	Number of simultaneously active queries	queries
c_x	Effective time for a named unit of work x	seconds/unit
$T_{\text{emb}}, Q_{\text{emb}}$	Query-embedding latency and its peak scratch RAM	seconds, bytes

Assume positive integer dimensions and counts, $1 \leq P \leq C$, $1 \leq K \leq R$, and a word width w divisible by eight. A selected pool may contain fewer than R eligible rows, in which case fewer candidates are returned.

Define two widths we will use repeatedly:

$$s_{\text{code}} = b_w \lceil d/w \rceil, \quad W_j = \lceil n_j/w \rceil, \quad s_{\text{mask},j} = b_w W_j. \quad (2)$$

The first describes one packed *document*; the second describes one bitplane across a *cluster*. They have different axes. If a filter leaves only a few eligible rows, W_j does not shrink unless the data is physically rebuilt or represented differently.

1.6 Three kinds of statement in this report

An algebraic identity, such as Equation (1), is exact under its stated score. An operation count is exact for the stated pseudocode or layout, but can change with the implementation. A latency or statistical estimate also needs hardware or distribution assumptions. Measured tables report a particular experiment. Keeping these categories separate is necessary for a fair comparison.

2 What is built once, and what every query pays

2.1 The public Exa-style starting point

Exa’s public description combines Matryoshka prefixes, binary document codes, float-query scoring through small lookup tables, similarity-based clustering, filtering, and final reranking [1]. Its published example uses particular dimensions and cluster counts; our formulas replace those choices with parameters. The pseudocode here is an explicit reference design, not a claim to reproduce Exa’s private implementation or production latency.

2.2 Preprocessing and the common index

A document is encoded before queries arrive. Its code and document ID are placed in its cluster’s contiguous row range. The baseline ownership convention is simple: each document has one packed code and one ID. Cluster offsets identify the ranges. Bitplanes or hash-key postings are additional structures described later.

Listing 1: Common preprocessing; executed before serving queries.

```
BuildCommonIndex(documents, model, d, routing_configuration):
    vectors = encode documents with model
    routing = train or choose the cluster assignment rule
    for each document x:
        b = signs of the first d coordinates of vectors[x]
        j = assign x to one cluster
        append (document_id[x], packed(b)) to cluster j
    concatenate cluster rows and record C + 1 offsets
    save centroids if centroid routing is used
    save filter metadata and optional final vectors
    return common index
```

Offline encoding, clustering and sorting are build costs. They are not charged as query latency. Their outputs *are* charged as index storage. Build-time temporary arrays can be large; the serving-RAM formulas below deliberately do not describe the construction peak.

For centroid routing, define

$$M_{\text{base}} = N s_{\text{code}} + N b_{\text{id}} + (C + 1) b_o + C d_r b_c + M_{\text{filter}} + M_{\text{extra}}. \quad (3)$$

M_{extra} includes explicitly chosen common metadata and optional stored final vectors. For example, fully resident reranking vectors contribute $N d_f b_f$. If they are on disk, they still consume storage, but only the resident portion belongs in serving RAM. For sign-prefix routing, omit the centroid term. Do not silently count a second ID array in one method and omit the only ID array in another.

Example: Stored bytes versus touched bytes

Suppose an illustrative database has $N = 10^9$ documents and $d = 256$ bits. Packed code payload without extra padding is $10^9 \cdot 256/8 = 32$ GB. With eight-byte IDs, the codes and IDs together need 40 GB before centroids or metadata. Here GB means 10^9 bytes. These are example choices for N, d, b_{id} , not constants in Equation (3).

2.3 Step 1: encode and prepare the query**Listing 2:** Query preparation shared by all methods.

```

PrepareQuery(text, model, D, d, d_r):
    full_query = model.encode(text, D)
    q = prepare prefix full_query[0:d] for the chosen score
    routing_query = prepare prefix full_query[0:d_r]
    t[i] = +1 if q[i] >= 0 else -1
    a[i] = abs(q[i])
    return full_query, q, routing_query, t, a

```

The model’s tokenization, neural-network computation and output processing cost $T_{\text{emb}}(\ell_q, D)$ for query length ℓ_q . This is a measured function, not a universal linear formula in D . Truncation, optional normalization, signs and magnitudes add $T_{\text{prep}} = O(D + d + d_r)$ elementary work. Their coefficient can be measured separately. A norm or layer-normalization rule must be the same rule used when preparing the documents.

Query-vector storage is $O(D + d + d_r)$ floats, with possible views instead of copies. Record the chosen ownership. We denote the actual retained query buffers by Q_{vec} and the embedding model’s own scratch peak by Q_{emb} .

2.4 Step 2: select clusters

With a flat centroid router, score all C centroids, then retain the best P . This is distinct from scoring binary documents: centroids are normally floats.

Listing 3: A simple flat centroid router with bounded selection memory.

```

Route(q_route, centroids, P):
    nearest = heap with capacity P
    for j in all C clusters:
        route_score = dot(q_route, centroid[j])
        offer (route_score, j) to nearest
    return the P selected cluster IDs

```

Let c_{fma} price one centroid coordinate’s multiply-add contribution, and let H_r be the number of heap comparisons or equivalent selection operations. Then

$$T_{\text{route}} = Cd_r c_{\text{fma}} + H_r c_{\text{sel}}, \quad H_r = O(C \log(P + 1)). \quad (4)$$

A streaming heap needs $O(P)$ records; materializing all centroid scores needs $O(C)$ scores instead. Either is valid, but their RAM formulas differ. Hierarchical or approximate routing has its own

measured visited-centroid count; replace C by that count only when the algorithm actually avoids the other centroids. Its routing recall also needs evaluation.

A sign-prefix router uses r fixed sign coordinates and at most 2^r addresses. It is not the same as similarity-based centroid clustering. Assigning a document is cheap, but ranking many possible query buckets is not free. One reference router scores all C bucket keys using $\lceil r/g \rceil$ lookup terms each and selects P , giving

$$T_{\text{route,sign}} = T_{\text{table,r}} + C \lceil r/g \rceil c_{\text{lut}} + H_r c_{\text{sel}}.$$

A direct or multi-probe address generator can replace that router, but must count its own key-generation and queue work. When every cluster is probed, routing can simply return all cluster IDs and skip distance-based selection.

2.5 Cluster count does not determine the selected document count

Exactly,

$$N_{\text{sel}}(q) = \sum_{j \in \mathcal{P}(q)} n_j. \quad (5)$$

Only under balanced clusters is this approximately NP/C . An optional load factor $\lambda(q) = N_{\text{sel}}/(NP/C)$ summarizes deviation for a *given query*; it is not automatically constant as P changes. For bitplanes, the individual n_j matter because wider clusters make each node more expensive. A mean load factor does not identify the largest cluster or peak frontier memory.

2.6 Step 3: apply optional filters

A filter can return eligible row IDs or a bitmap. Denote its time by T_{filter} and its scratch by Q_{filter} . For a bitmap intersection over the probed clusters, a reference work count is $\sum_{j \in \mathcal{P}} W_j$ word visits per input mask. For sorted ID lists, cost depends on list lengths and the intersection method instead.

After filtering, e_j rows are eligible in cluster j , so a scan scores $N_{\text{elig}} = \sum_j e_j$ documents. A dense bitplane still has physical width W_j . A filter that removes most rows therefore helps scoring immediately, but does not automatically make each bitplane intersection short.

The reference set for recall must obey the *same filter*. Otherwise a correct filtered search is unfairly penalized for not returning ineligible documents.

2.7 Step 4: build the float-query score tables

Split the d document signs into groups of g . A group has at most 2^g sign patterns. For each query group, precompute its dot product with every possible pattern.

Listing 4: The score table. Padding contributes zero in the last group.

```

BuildScoreTable(q, g):
    for group in ceil(len(q) / g) groups:
        for pattern in all g-bit patterns:
            table[group, pattern] = 0
            for coordinate in this group:
                add q[coordinate] * sign(pattern, coordinate)
    return table

ScoreDocument(packed_code, table):
    score = 0
    for group in all groups:
        pattern = extract this group's bits from packed_code
        score += table[group, pattern]
    return score

```

For this straightforward build,

$$G = \lceil d/g \rceil, \quad U_{\text{build}} \leq G2^g g, \quad (6)$$

$$T_{\text{table}} = U_{\text{build}} c_{\text{build}}, \quad Q_{\text{table}} = G2^g b_s, \quad (7)$$

$$T_{\text{score}}(F) = FGc_{\text{lut}}. \quad (8)$$

c_{lut} includes extracting the pattern, the lookup and accumulation in the chosen kernel. Faster table construction is possible, but it must replace U_{build} explicitly. Larger g reduces the number of lookup terms and increases the table exponentially. Register capacity and cache behavior therefore matter.

Example: Two query values and a four-entry table

For the first two values $(0.7, -0.4)$, the contributions for patterns 00, 01, 10, 11 are respectively $-0.3, -1.1, 1.1, 0.3$. With $g = 2$, document 1011 uses the first-group entry 1.1 and the second-group entry 0.1, giving 1.2.

Plain XOR followed by a population count computes *unweighted* Hamming distance. It matches Equation (1) only when all query magnitudes are equal up to a common scale. Replacing weighted scoring by that operation changes the task.

2.8 Step 5: retain candidates and optionally rerank

A top- R heap stores the worst retained candidate at its root. Every scored document needs at least a comparison with that root once the heap is full. Let $H_s(F, R)$ be the actual selection-operation count. A simple worst-case bound is $O(F \log(R + 1))$, not zero.

$$T_{\text{select}} = H_s(F, R)c_{\text{heap}}, \quad Q_{\text{results}} \approx R(b_{\text{id}} + b_s) + Q_{\text{heap,overhead}}. \quad (9)$$

For an optional rerank of R' returned candidates, $K \leq R' \leq R$, let

$$T_{\text{finish}}(R') = T_{\text{fetch}}(R', d_f) + R'd_f c_{\text{fine}} + H_f(R', K)c_{\text{fine,sel}} + T_{\text{output}}(K). \quad (10)$$

T_{fetch} depends on whether vectors are already resident, on SSD, or remote. If u_f vectors are fetched in a batch, the temporary float-vector buffer is $u_f d_f b_f$, not necessarily $R' d_f b_f$. The candidate IDs must remain live until fetching and scoring finish.

2.9 Shared latency and peak RAM

For sequential stages, define

$$T_{\text{common}} = T_{\text{emb}} + T_{\text{prep}} + T_{\text{route}} + T_{\text{filter}} + T_{\text{table}}. \quad (11)$$

The method-specific candidate work and Equation (10) are added later. For an index-only measurement, explicitly set the embedding and final-stage terms to zero.

Peak scratch RAM is a maximum of simultaneously live buffers, not a sum of every temporary array ever allocated. If embedding, search and reranking do not overlap,

$$Q_{\text{peak}} = \max\{Q_{\text{emb,stage}}, Q_{\text{search,stage}}, Q_{\text{finish,stage}}\}. \quad (12)$$

Each stage term includes buffers retained from previous stages. We use Q_{base} for common buffers live during candidate search, including query vectors, score tables, filter masks, routing output, and the candidate heap.

For J concurrent queries sharing one resident index,

$$\begin{aligned} M_{\text{serving}} &= M_{\text{resident}} + M_{\text{model}} + M_{\text{runtime}} + \max_t \sum_{u \in \mathcal{A}(t)} Q_u(t) \\ &\leq M_{\text{resident}} + M_{\text{model}} + M_{\text{runtime}} + JQ_{\text{peak,max}}. \end{aligned} \quad (13)$$

M_{model} counts shared resident embedding-model weights when inference is local. An external embedding service contributes latency but no local model-weight allocation. M_{runtime} is a chosen reserve or measured runtime overhead. It is not another stored copy of the index. Separate process replicas or device copies must be counted if the actual system has them.

For separate CPU and GPU memories, apply the residency and peak formulas to each memory pool. Count additional device copies if they exist. A throughput speedup does not by itself establish a feasible memory layout.

3 Approach A: cluster, then full scan

This is the reference candidate-search method. Once clusters and eligible rows are chosen, it scores every eligible row. There is no approximation *inside* that selected pool. The work is regular and can read packed codes sequentially when filtering does not scatter them.

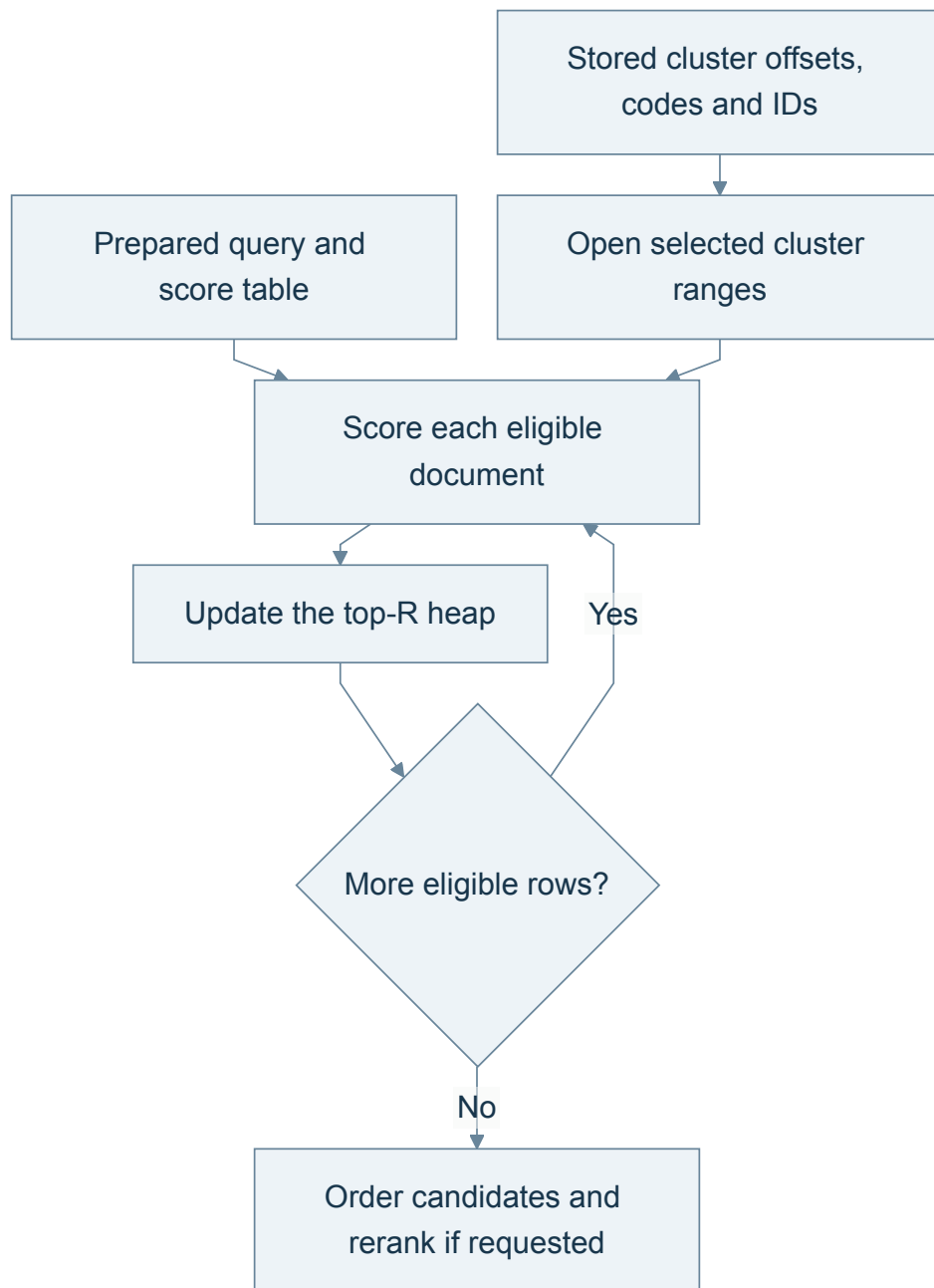


Figure 2: Approach A. The selected pool is smaller than the corpus only because of routing or filters.

3.1 Data layout and preprocessing

The common index from Equation (3) is sufficient. Cluster j occupies the row range between offsets j and $j + 1$. A row contains its packed code, and the corresponding ID array maps the row back to the original document.

Listing 5: Candidate search for Approach A.

```
ScanSelectedClusters(q, selected_clusters, eligible, table, R):
    best = empty top-R heap
    for cluster j in selected_clusters:
        for row in eligible rows of j:
            score = ScoreDocument(code[row], table)
            best.offer(score, document_id[row])
    return best in descending score, breaking ties by ID
```

The loop is over N_{elig} rows, not necessarily NP/C . If all clusters are searched and no filter removes documents, $N_{\text{elig}} = N$. Empty filtered clusters require no document scoring, although finding that they are empty can still have a cost.

3.2 Step-by-step cost

Step	Work	Live temporary data
Locate cluster ranges	P offset/range accesses	selected cluster IDs
Visit eligible rows	N_{elig} row iterations	current row/iterator
Score rows	$N_{\text{elig}}G$ lookup terms	shared score table, accumulator
Keep best R	$H_s(N_{\text{elig}}, R)$ selection operations	top- R heap
Return candidates	sort at most R results if needed	result array or heap storage

Writing c_{range} for range setup and $T_{\text{rank}}(R)$ for final candidate ordering,

$$T_{A,\text{candidates}} = Pc_{\text{range}} + N_{\text{elig}}Gc_{\text{lut}} + H_s(N_{\text{elig}}, R)c_{\text{heap}} + T_{\text{rank}}(R). \quad (14)$$

The general end-to-end latency is

$$T_A = T_{\text{common}} + T_{A,\text{candidates}} + T_{\text{finish}}(R'). \quad (15)$$

For the stated heap implementation, the candidate-stage time complexity is

$$O(P + N_{\text{elig}} \lceil d/g \rceil + N_{\text{elig}} \log(R + 1) + R \log(R + 1)).$$

The lookup term count is not a claim that each term is exactly one CPU instruction.

3.3 What is read, and what must remain in RAM?

The packed-code payload read for scoring is

$$V_{A,\text{codes}} = N_{\text{elig}} s_{\text{code}}. \quad (16)$$

ID reads, filter data, table accesses and result-heap accesses are additional traffic. A cache may satisfy many of them. Thus this is a named payload count, not a complete DRAM measurement. Sequential and scattered rows may have different effective c_{lut} or c_{row} .

Index storage and candidate-stage scratch are

$$M_{A,\text{index}} = M_{\text{base}}, \quad (17)$$

$$Q_{A,\text{search}} = Q_{\text{base}} + Q_{\text{row iterator}} + Q_{\text{scan workspace}}. \quad (18)$$

A streaming scan does not need an N -element score array. A matrix-based batch implementation may choose to allocate such an array; then that allocation must be added. The same algorithmic ranking can therefore have different peak RAM under different implementations.

3.4 A routing miss in the four-bit example

Partition the eight documents from Section 1 into

$$\mathcal{C}_0 = \{1, 3, 4, 6\}, \quad \mathcal{C}_1 = \{2, 5, 7, 8\}.$$

This is an illustrative fixed partition, not a claim about the output of a particular clustering run. Suppose the router opens only $\mathcal{C}_1$. A scans scores 1.2, 0.6, 0.8, -0.2 and returns documents 2 and 7. The global top two are 1 and 2, so recall is $1/2$. The scan did nothing wrong inside the opened cluster: document 1 was never considered.

Opening both clusters returns documents 1 and 2. This is why a scan can be exact in its selected pool and still have imperfect global recall.

3.5 SIMD changes a coefficient, not the score

SIMD means one instruction operates on several data lanes. It can accelerate code extraction, lookup accumulation, and selection. It does not by itself change the number of stored bits. There are two useful ways to model its effect.

Calibrated model. Measure the implemented kernel's effective time per work unit, c_{lut} , for the relevant dimensions and access pattern. If that rate already includes selection, use a combined $T_{\text{score+select}}$ and do not add the heap cost a second time.

Resource model. If a kernel performs U_{op} compute units and causes V_{DRAM} bytes of DRAM traffic, compute throughput Γ and memory bandwidth β give the lower bound

$$T_{\text{kernel}} \geq \max\{U_{\text{op}}/\Gamma, V_{\text{DRAM}}/\beta\}. \quad (19)$$

An effective SIMD factor s_{simd} can change Γ to $s_{\text{simd}}\Gamma_0$. It cannot make the bandwidth term disappear. Random access, synchronization and fixed overhead can put actual time above the bound. Do not add this resource estimate on top of a fitted coefficient that already includes the same work.

For multiple queries, a float score matrix can be written as $QB^\top$, where rows of B are expanded binary codes. That exposes matrix operations, but expansion, batching delay, device transfers and score-buffer storage have costs. A fast GPU batch is not automatically a fast single-query request.

What Approach A guarantees

For a fixed eligible pool, A returns its exact binary top R under the stated tie rule. With every eligible cluster searched, it has 100% binary recall. With partial routing, its global binary recall is limited by which true neighbors reach the selected pool. No property of the scan restores documents missed by routing, quantization or an earlier filter.

4 Approach B: cluster, then inverted bitplanes and branching

Approach B keeps the same document score and common routing stage. Its extra index stores the document bits by coordinate instead of only by row. That makes one sign test applicable to many document positions at once.

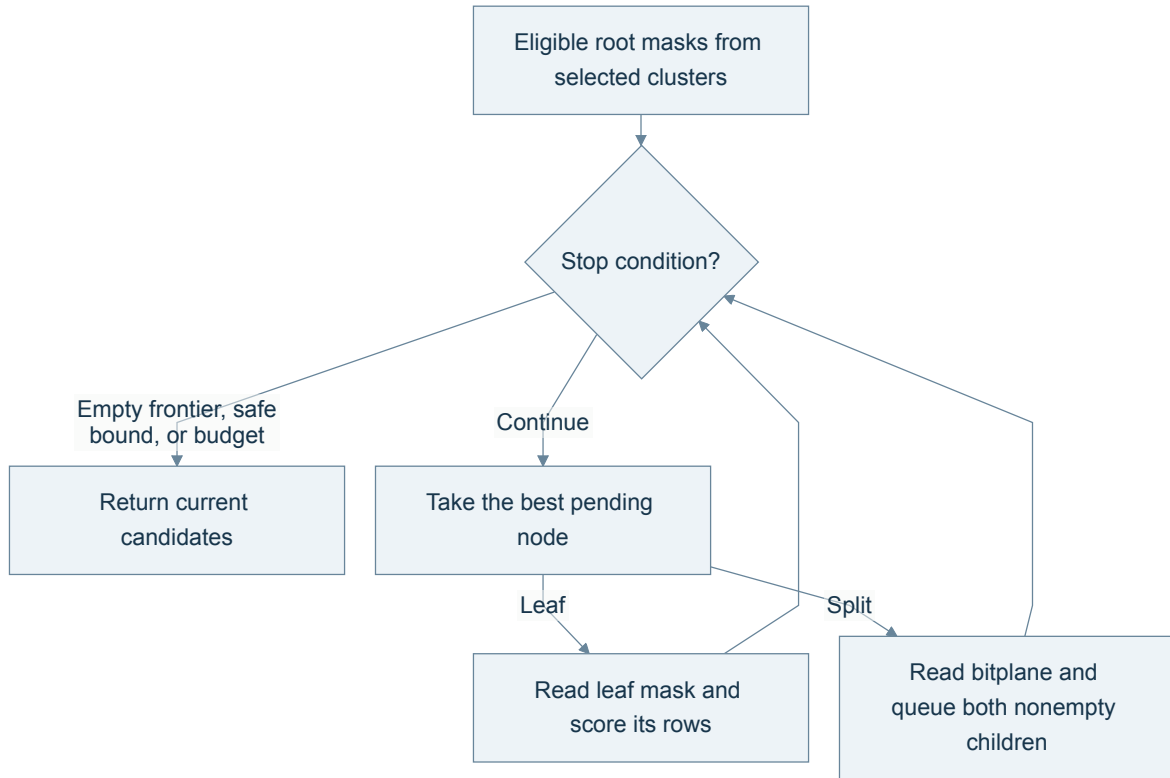


Figure 3: Approach B. A pending node has a set of active documents and an optimistic score bound.

4.1 What a bitplane stores

For the opened example cluster $\mathcal{C}_1 = (2, 5, 7, 8)$, the document codes are 1011, 1110, 1001, 0011. The transposed index is

	doc 2	doc 5	doc 7	doc 8
bitplane 1	1	1	1	0
bitplane 2	0	1	0	0
bitplane 3	1	1	0	1
bitplane 4	1	0	1	1

Each row is a bitmap over document positions, not a list of full document vectors. An active bitmap says which document positions still belong to the current node. To require coordinate i to be 1, intersect the active bitmap with plane i . To require 0, intersect with its complement, while masking unused padding positions.

Listing 6: Build the additional bitplane index.

```
BuildBitplanes(cluster_codes, d, w):
    for each cluster j:
        W[j] = ceil(number_of_rows(j) / w)
        planes[j] = d arrays of W[j] zero words
        for local row u in j:
            for coordinate i in 0 .. d-1:
                if code[j, u, i] is 1:
                    set bit (u mod w) in planes[j, i, floor(u/w)]
    return planes
```

The extra persistent storage is exactly, for this layout,

$$M_{\text{planes}} = db_w \sum_{j=1}^C W_j, \quad M_{B,\text{index}} = M_{\text{base}} + M_{\text{planes}}. \quad (20)$$

Without per-cluster padding this is $Nd/8$, a second binary-code-sized payload. For example, ten billion 256-bit codes occupy 320 GB; their bitplanes add approximately another 320 GB, not 32 TB. Padding and other index structures are additional.

4.2 Why the query suggests a traversal direction

At coordinate i , choosing the query's sign adds no penalty; choosing the opposite sign adds $a_i = |q_i|$. Sorting coordinates by descending a_i tests the largest potential penalties first. For the four-bit query, the order is 1, 2, 3, 4 with weights 0.7, 0.4, 0.2, 0.1.

In $\mathcal{C}_1$, matching bit 1 keeps documents 2, 5, 7; the alternative contains document 8 with accumulated penalty 0.7. Matching bit 2 next keeps 2, 7; document 5 moves to another pending node with penalty 0.4. This is a useful order, but neither alternative is automatically safe to discard.

Example: A greedy path can miss the best document

Take $q = (0.6, 0.5, 0.4, 0.3)$ and two documents: 0111 and 1000. The first has penalty 0.6 and score 0.6. The second has penalty 1.2 and score -0.6 . A greedy choice of the matching first bit selects 1000 and misses the better document. The largest individual weight is smaller than the sum of several remaining weights.

4.3 A bound makes branching safe

Let a node have accumulated penalty p on coordinates already tested. Untested coordinates cannot contribute a negative mismatch penalty. Consequently,

$$U(\text{node}) = A - 2p \quad (21)$$

is an upper bound on every active document's final score. If the result heap is full and U is strictly below its worst retained score, that node cannot improve the result. For equal scores, keep searching unless the ID tie rule is also certified.

The **frontier** is the collection of pending nodes. A best-first search takes the node with the largest upper bound, or equivalently the smallest penalty. The bound is optimistic: it may describe a sign combination that no actual document can achieve. A weak bound can keep many nodes alive even though none ultimately improves the answer.

Listing 7: Dense bitplane search across the selected clusters.

```
SearchBitplanes(q, selected_clusters, eligible, table, R, L, B_node):
    order = coordinates sorted by decreasing abs(q[i])
    frontier = empty min-penalty priority queue
    best = empty top-R heap
    for each selected cluster j:
        active = bitmap of eligible rows, with physical width W[j]
        if active is not empty:
            frontier.push(j, depth=0, penalty=0, active)
    visits = 0
    while frontier is not empty:
        if best is full and best_possible(frontier) < best.worst_score:
            break
        if a finite B_node is set and visits == B_node:
            break
        node = frontier.pop_smallest_penalty()
        visits += 1
        if node.count <= L or node.depth == d:
            for word in node.active:                # includes zero words
                for each set bit in word:
                    score its document and offer it to best
            continue
        i = order[node.depth]
        matching = active AND preferred_sign_plane(j, i, q)
        opposite = active AND NOT preferred_sign_plane(j, i, q)
        queue nonempty matching child with the same penalty
        queue nonempty opposite child with penalty + abs(q[i])
    return best
```

L is the leaf-size threshold and B_{node} the total node budget across all probes. An unlimited budget permits exact termination by bounds; a finite budget can stop before the remaining nodes are safely ruled out. A random exploration probability changes which pending node is visited next. It is not a direct formula for recall or a synonym for the number of nodes.

4.4 Count split visits, leaf visits, and scored documents separately

For each selected cluster, let

s_j = visited split nodes, ℓ_j = visited leaf nodes, f_j = documents fully scored.

Define $V = \sum_j (s_j + \ell_j)$ and $F = \sum_j f_j$. For a disjoint traversal without rescoring, $F \leq N_{\text{elig}}$. A finite node budget gives $V \leq B_{\text{node}}$, but does not determine F . The two principal bitmap counts are

$$U_{\text{split}} = \sum_j s_j W_j, \quad U_{\text{leaf}} = \sum_j \ell_j W_j. \quad (22)$$

A split word in the reference implementation reads an active word and a plane word, produces two child words, and counts matches. It includes two ANDs, a population count, complements and an accumulation. Allocating zero-filled child arrays can add two initialization writes. A leaf still scans all words in its active bitmap before enumerating set bits.

With separate effective coefficients,

$$T_{B,\text{candidates}} = T_{\text{order}} + T_{\text{roots}} + U_{\text{split}} c_{\text{split}} + U_{\text{leaf}} c_{\text{leafword}} + V c_{\text{node}} + FG c_{\text{lut,leaf}} + H_s(F, R) c_{\text{cheap}} + T_{\text{rank}}(R), \quad (23)$$

$$T_B = T_{\text{common}} + T_{B,\text{candidates}} + T_{\text{finish}}(R'). \quad (24)$$

Here $T_{\text{order}} = O(d \log d)$ and $T_{\text{roots}} = O(\sum_j W_j)$ for dense root construction. c_{node} may be replaced by explicit queue-comparison, mask-allocation and copy terms. Use one convention consistently; do not charge an allocation twice.

4.5 The full-width cost is the important hidden term

Example: Few scores do not imply little work

In the saved one-million-document run, $w = 64$ gives $W = 15,625$. A node budget of 32,768 produced a mean 22,557.88 split visits and 10,210.12 leaf visits. Therefore

$$U_{\text{split}} = 352,466,875, \quad U_{\text{leaf}} = 159,533,125.$$

Their sum is 512 million word-loop visits. About 163,766 documents were fully scored. With $g = 4$, that is 10.48 million score-table terms, versus 64 million terms for the scan. The reduction in scoring was real, but it came with hundreds of millions of bitmap visits. Word visits and score-table terms are not equal-cost instructions.

If a tree were fully explored, perfectly balanced, and every internal node had two nonempty children, then with L documents per leaf there would be roughly n_j/L leaves and one fewer split. This gives a dense bitmap term of order

$$O\left(\sum_j \frac{n_j^2}{Lw}\right). \quad (25)$$

This is an explanatory idealization, not a predictor for real embeddings. Fixed-bit splits can have only one nonempty child. A finite traversal also leaves unvisited nodes. Thus the actual split count need not be half the visited-node count.

A single greedy path visits only about $\log_2(n_j/L)$ levels in a balanced example, but still pays a full W_j at each level and has no general recall guarantee. In the broad worst case, a fixed-coordinate trie can contain $O(n_j d)$ nonempty prefix nodes, making dense traversal work as large as $O(d \sum_j n_j^2/w)$ before scoring and queue costs. Growing N alone does not guarantee that branching will overtake a scan.

4.6 Peak query RAM: the pending masks matter

Let $\mathcal{F}(t)$ be the frontier at time t , and let $j(v)$ identify a node's cluster. If a node record needs b_{node} bytes and bitmap capacity is fully allocated,

$$Q_{B,\text{frontier}}(t) = \sum_{v \in \mathcal{F}(t)} (b_{\text{node}} + b_w W_{j(v)}). \quad (26)$$

Capacity growth and allocator overhead can be represented by a factor $\gamma_B \geq 1$ or, preferably, by measured capacities. During a split, the parent and newly allocated children can coexist. A conservative candidate-stage expression is

$$Q_{B,\text{search}} = Q_{\text{base}} + Q_{\text{order}} + \max_t [\gamma_B Q_{B,\text{frontier}}(t) + Q_{\text{current}}(t) + Q_{\text{children}}(t)]. \quad (27)$$

For nonempty, disjoint pending sets, frontier count is at most N_{elig} . Starting with at most P roots, each processed binary split can increase that count by at most one, so a finite budget also gives $|\mathcal{F}| \leq P + B_{\text{node}}$. Multiplying that bound by the largest possible mask is conservative and can be very loose. It should not be reported as measured RAM.

When Approach B can help

B can help when it avoids enough full scoring to repay the bitmap, queue and allocation work. Useful conditions include strong early query penalties, small physical bitmaps, tight score bounds and a modest live frontier. Unlimited search is exact inside the selected pool under valid bounds and ties. Budget-limited search needs measured recall; its budget is not itself an accuracy percentage.

5 Approach C: cluster, then walk backwards from the ideal key

Approach C changes the lookup structure. Instead of repeatedly intersecting document masks, it asks: *which stored documents have this particular key?* Start from the signs the query prefers, then consider other keys in increasing mismatch penalty.

The primary method here still starts with the same selected clusters. A global hash table is the special case $C = P = 1$. We do not silently switch between clustered and global formulas.

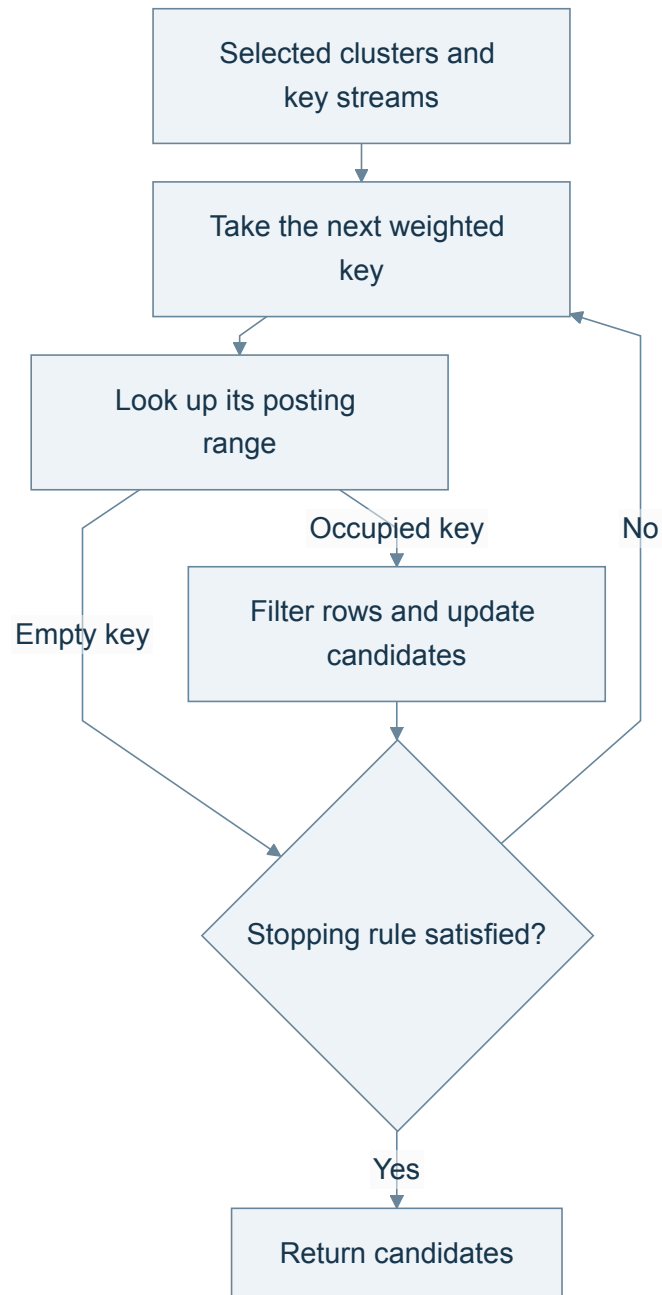


Figure 4: Approach C. Each cluster has a key directory and postings. Empty lookups still consume work.

5.1 Full keys and short keys are different choices

Let I_h be the fixed set of h lookup coordinates, often the first h positions. A **full key** has $h = d$. Its weighted penalty determines the complete binary score. A **short key** has $h < d$. It ranks a prefix, then the full d -bit score ranks the retrieved candidates.

A full-key search can be exact if it enumerates keys in true query-weighted order and finishes the result-boundary tie. A short-key candidate quota does not have that guarantee. It can still work well, and can also be made exact with a valid full-score stopping bound. These are properties of the stopping rule, not of the name “walk backwards.”

5.2 The data structure behind a key lookup

For each cluster, a directory maps a key to a posting range containing the local row references of documents with that key. There may be many documents per key. Finding a directory entry without finding its document rows is not enough to produce results.

Listing 8: Build a clustered key index.

```
BuildKeyIndex(cluster_codes, lookup_coordinates):
    for each cluster j:
        groups = group local row references by their lookup key
        concatenate the groups into a posting array
        for each occupied key:
            directory[j][key] = (posting_start, posting_count)
    return directories, postings
```

A **dense directory** has an offset for every possible key. A **sparse hash table** only stores occupied keys but adds key, slot and load-factor overhead. A sorted-key array is another option: it can use less overhead, but a lookup uses comparisons or another search structure instead of an assumed constant-time hash access.

5.3 Do not index fixed routing bits as if they were still free

If sign-prefix routing fixes some lookup coordinates inside cluster j , those coordinates cannot vary within it. Let r_j be the number of lookup coordinates fixed by that cluster, and let

$$h_j = h - r_j, \quad U_j \leq \min(n_j, 2^{h_j}). \quad (28)$$

U_j is the number of occupied free keys. For first- r -bit routing and a first- h -bit lookup with $h \geq r$, $h_j = h - r$. The fixed coordinates add a *constant* penalty for that cluster; they do not create 2^r additional keys inside it. Ordinary k-means clusters do not generally fix any sign coordinate, so this simplification does not follow from their cluster ID.

Example: Two fixed bits and a two-bit local key

Suppose a four-bit code is routed by its first two bits. Inside cluster 10, codes 1010, 1011, 1000, 1001 need only the local keys 10, 11, 00, 01. A lookup for the full prefix must include the cluster’s constant routing penalty when it is compared with a different cluster’s stream. The number of local possibilities is four, not sixteen.

5.4 Index storage: name every array

Use the common baseline row order and ID mapping. Let b_{ref} be the bytes for a local posting reference, $b_k(h_j)$ the bytes storing a key, and b_{slot} the remaining per-slot metadata. With a chosen hash load factor $0 < \alpha < 1$, a reference sparse-key layout uses

$$M_{\text{keydir}} = \sum_j \lceil U_j / \alpha \rceil [b_k(h_j) + 2b_o + b_{\text{slot}}], \quad (29)$$

where the two offsets represent start and length. Field alignment or different slot layouts can be substituted explicitly. The additional posting references cost Nb_{ref} . Therefore

$$M_{C,\text{index}} = M_{\text{base}} + Nb_{\text{ref}} + M_{\text{keydir}}. \quad (30)$$

This chosen layout deliberately counts both the common external-ID mapping and the extra row-reference permutation. If the build reorders the actual codes and IDs by key within each cluster, postings can become direct contiguous row ranges. Then the extra Nb_{ref} can be removed. That is an explicit layout change, not permission to omit all document IDs.

For a dense offset directory instead of a hash table,

$$M_{\text{dense-dir}} = \sum_j (2^{h_j} + 1)b_o.$$

Exponential key space makes dense directories expensive. Sparse directories avoid storing empty keys, but do not make probing empty keys free.

5.5 How to enumerate all flip combinations in weighted order

Let the free-coordinate weights be sorted in ascending order, $a_{(1)} \leq \dots \leq a_{(h_j)}$. Flipping a subset S costs $\sum_{i \in S} a_{(i)}$. Merely flipping the first bit, then the first two, then the first three misses other subsets. We need a stream containing *all* subsets in order.

The following priority-queue construction has at most two children per nonempty subset. It starts with the empty subset, then the subset containing coordinate 1.

Listing 9: A complete sorted-subset stream, not one chain of flips.

```
SortedFlipSets(weights):
    sort weights ascending, remembering original coordinate IDs
    emit (empty subset, penalty 0)
    if there are no free coordinates: stop
    queue.push(subset={1}, last=1, cost=weights[1])
    while queue is not empty:
        (S, i, cost) = queue.pop_smallest_cost()
        emit (S, cost)
        if i < number_of_weights:
            queue.push(S union {i+1}, i+1, cost + weights[i+1])
            queue.push((S minus {i}) union {i+1}, i+1,
                        cost - weights[i] + weights[i+1])
```

Both child costs are no smaller than the parent cost. Every nonempty subset has one parent: if the coordinate before its largest coordinate is also present, remove the largest one; otherwise

replace the largest coordinate by its predecessor. This explains why the construction visits every subset without needing a duplicate-state hash set.

A state must store enough information to construct the key. A full bitmask representation costs $b_w \lceil h_j/w \rceil$ bytes per state, plus its penalty, last coordinate and queue metadata. Copying that mask into children is work. A persistent representation can change this cost, but it then needs its own traversal and memory accounting.

Example: The full-key order can be accurate even when a greedy branch is not

For $q = (0.6, 0.5, 0.4, 0.3)$, the ideal code is 1111. The first penalties include 0, 0.3, 0.4, 0.5, 0.6, 0.7, and so on. The code 0111 is found by flipping coordinate 1 at penalty 0.6. The code 1000 has penalty 1.2 and comes later. Complete weighted enumeration finds the better document first, even though it disagrees on the query's largest coordinate. It may have to check several empty keys before that hit.

5.6 Lookup and stopping within selected clusters

For short keys, a simple implementation allocates candidate targets m_j to probed clusters. A more flexible implementation merges their next-prefix-key scores and draws candidates from the best available stream. Whichever policy is used must be a parameter of the experiment; a per-cluster target and a global target are not interchangeable.

Listing 10: Short-key candidate search with explicit per-cluster targets.

```
SearchShortKeys(q, selected_clusters, h, targets m[j], R):
    candidates = empty collection
    for each selected cluster j:
        stream = SortedFlipSets(free lookup weights in j)
        found = 0
        for (flip_set, variable_penalty) in stream:
            key = ideal_free_key(q, j) XOR flip_set
            rows = directory[j].lookup(key)      # may be empty
            keep eligible rows from this posting range
            found += number of kept rows
            append those rows to candidates
            if found >= m[j] and final prefix-penalty ties are finished:
                break
    score every candidate with the full d-bit score
    keep the best R with the same score/ID rule as A
    return best R
```

A full-key exact version instead merges the cluster streams by their *total full-key* penalty, including fixed-sign penalties. Once at least R documents have been found, continue through every key that can tie or exceed the R th document. This produces the exact binary top R within the selected pool. If unprobed clusters remain, global exactness additionally requires their score bounds to be ruled out or those clusters to be searched.

A short key can also stop exactly. Let $T_h(b)$ be the prefix score and $A_{\text{tail}} = \sum_{i \notin I_h} |q_i|$. Every document in an unvisited prefix key satisfies

$$S(q, b) \leq T_h(b) + A_{\text{tail}}. \quad (31)$$

If the best unvisited prefix upper bound is strictly below the current worst top- R score, the result is certified. If not, continue even when a nominal candidate quota has been reached. A larger tail makes this bound less selective. Fixed routing-coordinate contributions must be included consistently; no coordinate may be counted twice.

5.7 Do we need to rescore a document whose full key is known?

Not always. With a full key and the true query weights, its complete score is already $A - 2\pi_q(\text{key})$. Every document in that key’s posting range has that same binary score. The search can offer their IDs to the result heap without unpacking and rescoring each code. For a short key, the known prefix contribution can similarly be reused while evaluating only the remaining coordinates.

To make that implementation choice visible, let $G_{C,j}$ be the number of score-table groups actually evaluated per candidate in cluster j . The simple pseudocode above has $G_{C,j} = G$. A full-key implementation reusing its exact key score can have $G_{C,j} = 0$. A tail-only scorer can have approximately $\lceil d_{\text{remaining}}/g \rceil$ groups, with alignment and fixed routing coordinates handled explicitly. It still pays for postings, key-score arithmetic, selection, and any later float reranking. This optimization changes the cost; it is not a recall loss.

The reference storage formula retains the common packed rows. A specialized full-key layout can also avoid a redundant per-document code copy by using its stored full keys, provided its ID/posting and key ownership are redesigned and counted. Equation (30) is one explicit layout, not a universal lower bound on every possible key index.

5.8 If “work upwards” means removing a constraint

There is another reasonable interpretation of the idea. Instead of flipping a bit and probing one more exact key, remove a bit constraint. For example, $101*$ means either 1010 or 1011 . That is a set of keys. A plain exact-key hash table does not answer this wildcard request in one ordinary lookup.

A fixed-order **prefix trie** stores paths by successive key coordinates. A node identifies an occupied prefix. If document rows are ordered by key within a cluster, each node can store the start and end of a contiguous posting range. Moving to its parent relaxes the last fixed coordinate. It does not require copying every descendant’s posting list into every ancestor.

Example: Relaxing the key inside the example cluster

For $\mathcal{C}_1 = \{2, 5, 7, 8\}$ and ideal key 1010:

Constraint	Matching documents	Newly visited documents
1010	none	none
101*	2	2
10**	2, 7	7
1***	2, 5, 7	5
****	2, 5, 7, 8	8

The parent range contains the previous range. Scan only the newly added sibling ranges if the implementation can retain their boundaries. Otherwise repeated scans must be counted again. In this particular query the last coordinate is the cheapest one to relax. That is not true for every query.

Listing 11: A fixed-prefix relaxation variant of Approach C.

```

SearchByRelaxingPrefix(q, query_key, trie, table, R, candidate_target):
    path = follow query_key as far as the stored trie permits
    previous_range = empty
    best = empty top-R heap
    found = 0
    for node on that path, from deepest node toward the root:
        new_ranges = node.posting_range minus previous_range
        for eligible row in new_ranges:
            score its code and offer it to best
            found += 1
        previous_range = node.posting_range
        if found >= candidate_target:
            return best                                # approximate stopping rule
    return best

```

An exact version may stop early only after bounding *every* unvisited region against the current full-score heap. Simply finding enough IDs in a relaxed prefix is not such a proof. The remaining score can still change the ordering.

Let Z_j be the number of stored trie nodes and b_{trie} their record size, including child links or equivalent navigation and range offsets. A simple uncompressed trie satisfies

$$Z_j \leq 1 + \sum_{u=1}^{h_j} \min(n_j, 2^u).$$

Its directory term replaces the hash-directory term by $\sum_j Z_j b_{\text{trie}}$. Path compression can reduce the actual node count. With range references, document postings are stored once; copying a posting list into every prefix node would instead add repeated references and can cost $O(Nh)$ reference storage.

For u_j ancestor/range visits and F_j total row visits, a reference cost is

$$T_{C,\text{trie}} = T_{\text{keyprep}} + \sum_j [h_j c_{\text{path}} + u_j c_{\text{range}} + F_j c_{\text{post}} + F_j G_{C,j} c_{\text{lut,gather}}] + T_{\text{select/rank}}. \quad (32)$$

A saved navigation path uses $O(h_j)$ records per active stream. Add the candidate heap, gather buffers, and common query RAM. Here F_j must count repeated rows if they are revisited. The same common pipeline and final rerank terms are added for end-to-end latency.

A fixed trie can drop its last prefix coordinate cheaply. Dropping an arbitrary query-selected low-magnitude coordinate can produce several disjoint ranges, so it does not have the same constant-cost parent operation. A query-adaptive bit order requires additional traversal or a different index. This is the main difference between a prebuilt fixed trie and the query-ordered bitplane traversal.

5.9 Count keys, postings, scores and enumeration states

Let H_j be the number of keys attempted in cluster j , including empty keys. Let F_j be the number of eligible posting rows retrieved and scored. Let E_j be the number of sorted-subset heap comparisons, C_j^{copy} the number of key-word copies, and let Q_j^{max} be the largest enumeration-queue size. Define $F = \sum_j F_j$.

For the two-child stream, $E_j = O(H_j \log(Q_j^{\text{max}} + 1))$ comparison work, and $C_j^{\text{copy}} = O(H_j \lceil h_j/w \rceil)$ for full-mask copies. Hashing a key also depends on its length and table layout. An effective $c_{\text{lookup}}(h_j, \alpha)$ should price the emitted-key iteration outside heap comparisons and child-mask copies: key construction, fixed state bookkeeping, the directory lookup, and obtaining the current key score or bound. This includes cache effects and collisions, rather than assume one CPU operation.

$$T_{C,\text{candidates}} = T_{\text{keyprep}} + \sum_{j \in \mathcal{P}} \left[E_j c_{\text{enum}} + C_j^{\text{copy}} c_{\text{copy}} + H_j c_{\text{lookup}}(h_j, \alpha) + F_j c_{\text{post}} \right] \\ + \sum_j F_j G_{C,j} c_{\text{lut,gather}} + H_s(F, R) c_{\text{cheap}} + T_{\text{rank}}(R), \quad (33)$$

$$T_C = T_{\text{common}} + T_{C,\text{candidates}} + T_{\text{finish}}(R'). \quad (34)$$

T_{keyprep} includes sorting lookup weights, initializing streams and constructing query keys. For P different free-coordinate sets, a straightforward upper count is $O(\sum_j h_j \log(h_j + 1))$. A global merge adds its own P -stream selection work, typically $O((\sum_j H_j) \log(P + 1))$; include it when that variant is used.

The hash directory can be built before the query, but query-specific weighted order depends on q . Precomputing a hash table does not precompute this order for every float query.

5.10 Query RAM while enumerating

Let $b_{\text{enum},j}$ include a key mask, penalty and queue metadata. For simultaneously active streams, the candidate-search scratch is

$$Q_{C,\text{search}} = Q_{\text{base}} + \max_t \left[\sum_j \gamma_C Q_j(t) b_{\text{enum},j} + Q_{\text{stream merge}}(t) + Q_{\text{posting buffer}}(t) + Q_{\text{gather}}(t) \right]. \quad (35)$$

A sequential per-cluster implementation can reuse enumeration memory, replacing the sum of live stream sizes by the maximum live cluster stream. A global full-score stream may need all selected streams active. These are different execution choices.

Postings can be streamed into a top- R heap; storing all F retrieved IDs is not mandatory. If they are materialized for batch reranking, add Fb_{ref} or the actual bounded batch size. A sparse dictionary's small stored size does not bound the size of a query's enumeration queue. With the two-child stream, the queue can grow by at most one per emitted state, but H_j itself can be enormous.

What Approach C does and does not lose

Full-key enumeration in true penalty order can have 100% binary recall. Its difficulty can be time or queue memory, especially when most keys are empty. Short-key search is a different tradeoff: it can collect useful candidates cheaply, but a fixed quota is not a recall guarantee. It becomes exact if a valid full-score bound proves that no unseen candidate can improve the result, or if it eventually scores every eligible document.

6 Deriving recall without assuming the answer

The work formulas require counts such as visited nodes, retrieved postings and scored rows. They do not, by themselves, give recall. Recall depends on *which* rows survive and on the reference ranking. This section starts with an exact definition, then adds assumptions one at a time so their consequences can be checked.

6.1 The exact measured definition

Let $\mathcal{E}(q)$ be the eligible document set after the intended filter, across the whole database. Let $\mathcal{G}_K(q)$ be its exact top K under the chosen reference score, and let $\mathcal{R}_K(q)$ be the returned results. Then

$$\text{Recall@ } K(q) = \frac{|\mathcal{G}_K(q) \cap \mathcal{R}_K(q)|}{K}. \quad (36)$$

Use the same tie rule in both sets. Average this quantity over queries. Repeating one query for timing does not create new independent quality observations.

If a candidate set $\mathcal{D}(q)$ is reranked with the *same exact reference score*, every true top- K document in that set survives into the final top K . Therefore

$$\text{Recall@ } K(q) = \frac{|\mathcal{G}_K(q) \cap \mathcal{D}(q)|}{K}. \quad (37)$$

A different reranking score breaks that identity. In particular, exact binary candidates may still miss the true float-vector top K or the documents relevant to a human question.

6.2 Separate routing loss from candidate-search loss

Let $\mathcal{U}(q)$ be the eligible pool in the probed clusters. Define

$$r_{\text{route}}(q) = \frac{|\mathcal{G}_K(q) \cap \mathcal{U}(q)|}{K}, \quad r_{\text{local}}(q) = \frac{|\mathcal{G}_K(q) \cap \mathcal{D}(q)|}{|\mathcal{G}_K(q) \cap \mathcal{U}(q)|}.$$

When the denominator is positive and $\mathcal{D} \subseteq \mathcal{U}$,

$$\text{Recall@ } K(q) = r_{\text{route}}(q)r_{\text{local}}(q). \quad (38)$$

If no true top- K result enters the selected pool, total recall is zero. This factorization is exact when both factors refer to the *same true global neighbors*. However, multiplying an average routing recall by an average local recall is generally wrong:

$$\mathbb{E}[r_{\text{route}}r_{\text{local}}] = \mathbb{E}[r_{\text{route}}]\mathbb{E}[r_{\text{local}}] + \text{Cov}(r_{\text{route}}, r_{\text{local}}).$$

Nor is recall against a cluster's own local top K automatically the same as recall for the particular global neighbors routed into it. That distinction matters when transferring the flat-pool branch measurements into a clustered model.

For A, local binary search is exact. For unlimited B with valid bounds, it is exact as well. For full-key C with complete weighted ordering, it is exact. Their global recall can still be limited by routing. Finite node budgets and short-key quotas add a second possible loss.

6.3 A deterministic error bound for a short prefix

Split the full score into retained and omitted coordinates:

$$S = T_h + Z_h, \quad T_h = \sum_{i \in I_h} q_i b_i, \quad Z_h = \sum_{i \notin I_h} q_i b_i. \quad (39)$$

Because $b_i \in \{-1, +1\}$,

$$|Z_h| \leq A_{\text{tail}} = \sum_{i \notin I_h} |q_i|. \quad (40)$$

Two documents can reverse their prefix order if the omitted contributions overcome their prefix score difference. A prefix gap greater than $2A_{\text{tail}}$ is sufficient to preserve that pair's order. This is a sufficient bound, not a typical-error estimate.

A more useful query-time certificate compares the best unseen prefix score plus A_{tail} with the worst fully scored retained result, as in Equation (31). It adapts to the actual query. If the bound is too loose, more keys must be visited. Returning a fixed candidate count without checking that condition remains approximate.

For float truncation, the analogous score error obeys Cauchy–Schwarz:

$$|q_{\text{tail}}^\top x_{\text{tail}}| \leq \|q_{\text{tail}}\|_2 \|x_{\text{tail}}\|_2.$$

Truncation and binary quantization are separate approximations. Matryoshka training can improve the usefulness of prefixes, but it does not set either bound to zero or supply a universal recall percentage.

6.4 Random documents: what the independence assumption actually gives

The following distribution models first omit routing. When routing is added, retain the same global true-neighbor event or apply Equation (38); replacing it with a different local top- K event changes the question. Assume nonzero score variance when computing a correlation.

First condition on a particular query q . Assume document signs are independent and equally likely to be -1 or $+1$. Then

$$\mathbb{E}[S \mid q] = 0, \quad \text{Var}(S \mid q) = \sum_{i=1}^d q_i^2, \quad (41)$$

$$\mathbb{E}[T_h \mid q] = 0, \quad \text{Var}(T_h \mid q) = \sum_{i \in I_h} q_i^2, \quad (42)$$

$$\text{Cov}(T_h, S \mid q) = \text{Var}(T_h \mid q). \quad (43)$$

Hence the score correlation is exactly

$$\rho_q = \text{Corr}(T_h, S \mid q) = \sqrt{\frac{\sum_{i \in I_h} q_i^2}{\sum_{i=1}^d q_i^2}}. \quad (44)$$

High retained variance is not the same as a small worst-case tail bound. Many small omitted coordinates can have a modest sum of squares but a large sum of absolute values. A useful approximate prefix can therefore still lack a cheap exactness certificate.

The square root is important. Retaining a fraction v of score variance gives correlation $\sqrt{v}$, not recall v and not recall $\sqrt{v}$.

Example: Two random-data cases can have very different prefix recall

With equal coordinate variances, a 24-coordinate prefix of a 256-coordinate query retains roughly 24/256 of expected score variance. If instead the first 24 coordinates are constructed to hold 80% of expected score variance, their score is much more strongly correlated with the full score. The controlled measurements in Section 7 test both cases. The second is a deliberately favorable distribution, not the definition of Matryoshka embeddings.

If query coordinates are random normals $q_i \sim \mathcal{N}(0, \sigma_i^2)$, we can choose a variance profile σ_i^2 . The ratio of *expected* prefix and full variances is $\sum_{i \in I_h} \sigma_i^2 / \sum_i \sigma_i^2$. For a specific sampled query, use its actual Equation (44). The expectation of a ratio need not equal the ratio of expectations.

6.5 Real documents need a covariance model, not just a label

A covariance matrix records how coordinates vary together. Its diagonal entries are individual variances. An off-diagonal entry describes whether two coordinates tend to move together or in opposite directions. Independent, balanced signs have unit diagonal entries and zero off-diagonal entries.

For a general binary document vector with mean μ and covariance matrix Σ , let q_h be q with omitted coordinates set to zero. Then

$$\rho_q = \frac{q_h^\top \Sigma q}{\sqrt{(q_h^\top \Sigma q_h)(q^\top \Sigma q)}}. \quad (45)$$

This includes dependencies between coordinates. Conditioning on a cluster or filter can change both μ and Σ . A covariance measured globally is not automatically the conditional covariance inside the query's selected clusters.

A learned prefix may preserve ranking information through these dependencies even when its raw coordinate-energy share is small. Conversely, a prefix can have considerable energy and still fail to identify the extreme nearest neighbors. Estimating covariance also does not prove a normal distribution in the tails.

6.6 Equal-weight mistakes lead to a binomial model

Let $X_i = \mathbf{1}[b_i \neq t_i]$. If the X_i are independent with the same disagreement probability p , and all relevant weights equal a , then a mismatch count over h coordinates satisfies

$$M_h = \sum_{i=1}^h X_i \sim \text{Binomial}(h, p), \quad \pi_h = aM_h. \quad (46)$$

A Hamming radius r visits

$$V_h(r) = \sum_{u=0}^r \binom{h}{u} \quad (47)$$

keys, and its expected candidate population is

$$N_* \Pr(M_h \leq r) = N_* \sum_{u=0}^r \binom{h}{u} p^u (1-p)^{h-u}. \quad (48)$$

Here N_* is the relevant reference pool size; it is N without filters or cluster restriction. Equation (48) is a candidate-count model, not yet recall.

If $p = 1/2$, every key has equal probability and the expected count reduces to $N_* V_h(r)/2^h$. Different p_i with equal weights give a Poisson-binomial count. Unequal weights give a weighted Bernoulli sum. Neither is described by one ordinary binomial with an average p .

6.7 Derive recall by asking which true neighbors pass

Write $F_X(x) = \Pr(X \leq x)$ for a cumulative distribution function (CDF): the probability that X is at or below a threshold.

Let A_h be the prefix penalty and B_h the omitted-coordinate penalty. Full penalty is $A_h + B_h$. Under coordinate independence, the two sums are independent. Let u define the true full-score tail, and let v be the candidate prefix-penalty threshold. Then

$$r_{\text{population}}(u, v) = \frac{\Pr(A_h \leq v, A_h + B_h \leq u)}{\Pr(A_h + B_h \leq u)} = \frac{\int_{a \leq v} F_{B_h}(u - a) dF_{A_h}(a)}{F_{A_h + B_h}(u)}. \quad (49)$$

Read the numerator as a weighted sum: for each allowed prefix penalty, ask whether the remaining coordinates fit within the full-score threshold, multiply by the probability of that prefix penalty, and add.

This is the logic missing from a candidate-count-only calculation: a retrieved document counts toward recall only if it also belongs to the true full-score tail.

For equal weights and an integer full radius r_d , the formula becomes

$$\frac{\sum_{u=0}^{\min(h, r_h, r_d)} \binom{h}{u} p^u (1-p)^{h-u} F_{\text{Bin}(d-h, p)}(r_d - u)}{F_{\text{Bin}(d, p)}(r_d)}. \quad (50)$$

This is exact for the stated population events. Choosing those radii to approximate top K and a candidate budget is another step. Finite-sample ranks and score ties need care.

Example: Candidate fraction is not recall

Use all 64 six-bit codes as documents and an all-positive equal-weight query. The best seven codes have zero or one mistake. Keep only codes matching the first two query bits: 16 codes pass, so the candidate fraction is $16/64 = 25\%$. Five of the true seven pass: the ideal code and the four codes whose one error is in an omitted coordinate. Recall is $5/7$, about 71.4%, not 25%. Allowing one error in the two-bit prefix retrieves 48 codes and includes all seven. Both numbers follow by counting the same true-neighbor event.

6.8 Unequal query weights: count weighted subsets

For a fixed query and lookup coordinates I_h , define

$$V_q(v) = \# \left\{ S \subseteq I_h : \sum_{i \in S} |q_i| \leq v \right\}. \quad (51)$$

This counts every attempted key through threshold v , whether occupied or empty. Under independent uniform keys, expected candidates are $N_* V_q(v)/2^h$. With real key probabilities,

$$\mathbb{E}[F(v) \mid q] = N_* \sum_{k: \pi_q(k) \leq v} \Pr(\text{key} = k \mid q, \text{cluster}, \text{filter}). \quad (52)$$

For a clustered lookup, apply Equation (52) separately with pool size e_j and the conditional key distribution in cluster j , then sum the results.

An average bit-disagreement rate is insufficient to determine these joint probabilities. The empty-key count V_q and occupied candidate count F are separate inputs to the latency formula. A dense document distribution near the query key can make F large while V_q remains small; a sparse key space can do the reverse.

For small h , split weights in two, enumerate each half's subset sums, sort one list, and count pairs whose sum is at most v . This meet-in-the-middle calculation uses roughly $O(2^{h/2})$ memory and $O(h2^{h/2})$ sorting/search work. It is an analysis tool for counting keys, not the measured latency of an online hash enumerator.

For quantized nonnegative integer weights a_i , a dynamic program computes probability mass:

$$f_i(s) = (1 - p_i) f_{i-1}(s) + p_i f_{i-1}(s - a_i). \quad (53)$$

Replacing probabilities by counts and adding both predecessor counts computes key volumes. This is exact for the quantized-weight model. It does not silently become an exact calculation for unquantized query weights. If score quantization incurs a uniform score error at most ϵ , a score gap greater than 2ϵ is sufficient to preserve a pair's order. Otherwise candidates near the boundary need exact scoring or a safe expanded bound.

6.9 The conditional-normal approximation

For many small independent contributions, a joint-normal approximation for full and prefix scores can be convenient. Subtract each score’s mean and divide by its standard deviation. This standardizes them to (Z, Y) with correlation ρ . For a true fraction $a = K/N_*$ and selected candidate fraction f , set

$$z_K = \Phi^{-1}(1 - a), \quad y_f = \Phi^{-1}(1 - f),$$

where Φ and ϕ are the standard normal CDF and density. The implied recall is

$$\hat{r} = \frac{1}{a} \int_{z_K}^{\infty} \phi(z) \Phi\left(\frac{\rho z - y_f}{\sqrt{1 - \rho^2}}\right) dz. \quad (54)$$

We average over the *entire* true tail. Substituting only $z = z_K$ calculates a boundary conditional probability, not the integral. When $\rho = 0$, recall is f . In the limit of identical rankings, $\rho = 1$, it is $\min(1, f/a)$, subject to ties.

This formula is not valid simply because an embedding is called Matryoshka. A short weighted binary prefix has finite, discrete support. Its extreme-score distribution can differ greatly from a normal tail. At a billion documents, top- K events may be far deeper in the tail than any small validation sample covers. Strong overall score correlation does not certify 99% recall.

The controlled study below fits score correlation on one query group and checks predicted recall on different queries. Even there, the real-data normal approximation misses by several percentage points. Such a model can generate hypotheses; it should not declare a real 99%-recall winner.

6.10 What can be concluded about the three methods

- **A:** exact local scanning has no local candidate-search loss; global loss comes from routing, candidate/reranker score changes, and the declared filtering task.
- **B:** exact bound-based traversal has the same binary result as A on the same selected pool. Finite budgets require query-level measurements or a distribution model for visited nodes and surviving true neighbors. A node count alone is not enough.
- **C:** full keys with complete weighted ordering are exact locally. Short keys can be accurate when their ranking preserves true neighbors and the candidate budget is sufficient. That possibility should be measured, not dismissed. Their enumeration and posting costs remain separate from the recall benefit.

The exact empirical definition is always available. Distribution models explain mechanisms; validation determines where their numerical estimates are useful.

7 What the existing measurements support

The purpose of these measurements is to check the reasoning, not to fit every equation until one method appears to win. The experiments have different scopes, so their numbers must not be merged into one unqualified performance claim.

7.1 The one-million-passage scaling study

The archived study used one flat cluster, a fixed 256-bit score, 100 binary candidates, leaf size 32, deterministic branching, cached query embeddings, and one CPU thread. There were 100 validation queries for selecting configurations and 200 held-out test queries, with three timing repeats. It covered pools from 1,000 to one million real MS MARCO passages.

One-million-document test	Median latency	Binary recall	Fully scored rows
Packed scan	25.14 ms	100%	1,000,000
Branch, budget 32,768	244.98 ms	99.26%	about 163,766
Unlimited branch	1,068.84 ms	100%	about 999,961

The branch reduced scoring at the finite budget, but did not win at matched target recall. The whole study found no median or p95 crossover among the tested 95% and 99% target settings. This conclusion belongs to that implementation, dataset, hardware and grid. It is not a theorem that every bitplane representation loses.

The original implementation uses dense active masks for every node. Its measured split-word counter omits leaf reads, which must be derived from the visited-node counts and cluster widths. The physical mask size at one million rows is $15,625 \times 8 = 125,000$ bytes. A frontier with 20,000 such masks would contain 2.5 GB of masks alone. This example explains how query RAM can be much larger than the packed-code payload; it is not a claimed observed frontier count.

The source summary is `results/scaling-msmarco-1m/analysis/settings.csv`. Full run metadata and the validation protocol live beside it. The report’s verification script checks the numerical rows quoted here.

7.2 The FiQA study asked a broader pipeline question

The denser FiQA run used 57,638 documents and 648 queries, with 583 settings including random seeds. Its timings include routing, binary candidate search and float reranking, with embeddings cached. Its quality measures include float Recall@10 and nDCG@10, rather than the scaling study’s binary Recall@100.

Method	Probes	Node budget	Median ms	Float Recall@10
Sign-route binary scan	16	–	0.7524	0.6784
Sign-route branch	32	2,048	0.6256	0.7006
Sign-route branch	64	2,048	0.6466	0.7062
Float IVF reference	8	–	0.1661	0.9127

These selected examples show that probing more clusters with limited work inside them can be a useful combined tradeoff. They do not isolate traversal efficiency: the selected pools differ. Within the same selected buckets, unlimited branching was slower than scanning. The float IVF row uses a different scorer and was a strong practical reference in that experiment.

Random exploration at probability 0.1 did not show a consistent improvement. Its purpose is to change visit order; it does not provide a mathematical guarantee of higher recall. The full FiQA report retains losing configurations, empty returns, and the relevant measurement scope. Its source is `results/fiqa-2026-09-09-dense/README.md` and the accompanying data.

7.3 A new controlled check of the 24-bit prefix

The small quality study used the same full weighted-binary score and row-ID tie rule for all methods. There was no outer routing: $C = P = 1$. Two synthetic datasets had 20,000 random binary documents. The real-data case used the first 100,000 rows of the cached Nomic/MS MARCO sample. All cases used $d = 256$ and $h = 24$.

One synthetic query distribution gave all coordinates equal variance. The other multiplied the first 24 coordinates so that they held 80% of *expected* full-score variance. It was a controlled favorable case for a prefix, not a model of what Matryoshka always does.

For each case, 16 queries estimated a score correlation and 16 different queries checked the normal-model prediction. The native full scan agreed with an independent exact score sort on all 96 query references. The short-key candidate set included all documents in keys tied at the final prefix threshold, then used the exact full score to keep the top 100.

Data, candidate target	Measured recall	Normal prediction	Gap	Median keys
Random equal variance, 1,000	23.00%	21.84%	−1.16 pp	846,156
Strong prefix, 1,000	95.00%	97.01%	+2.01 pp	846,156
Cached Nomic, 1,000	25.25%	17.41%	−7.84 pp	6,976
Cached Nomic, 5,000	52.13%	42.57%	−9.56 pp	35,504
Cached Nomic, 10,000	66.19%	58.46%	−7.73 pp	78,370

“Gap” is prediction minus measurement in percentage points. Values are rounded from the per-query evidence. The identical synthetic key counts are expected: the strong-prefix case scales all first-24 query weights by the same factor, which changes neither their key order nor the prefix candidate sets. It changes their importance in the *full* score.

These results support two different conclusions. First, a short prefix is not inherently inaccurate: the constructed favorable case reaches 95% recall with 1,000 candidates. Second, a simple statistical prediction can still miss real recall by several points. An error of that size is unacceptable as a certificate of a 99% recall target.

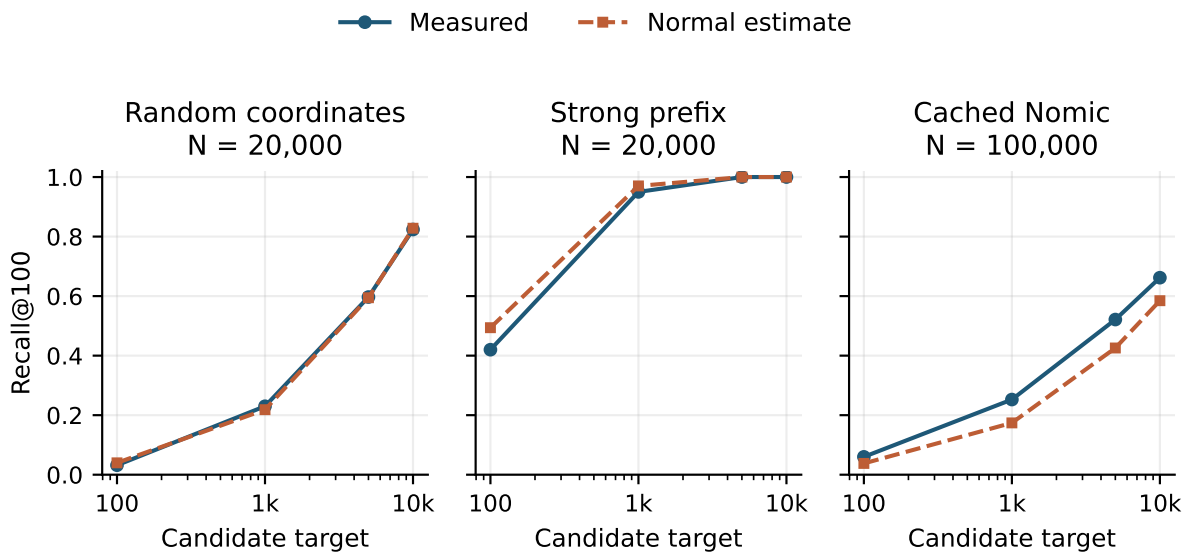


Figure 5: Measured short-key recall versus the checked normal approximation. Each panel has its own explicitly stated document population.

The key counts were calculated from the actual query weights by meet-in-the-middle counting, including empty keys. The candidate set was constructed using an exhaustive prefix-score calculation for quality evaluation. **That computation is not an implemented hash-search latency benchmark.** Its wall time must not be plotted as the latency of Approach C.

In the same favorable synthetic case, native B at budget 512 recovered about 99.38% of the binary top 100 while fully scoring about 4,122 documents on average. In the real-data case, B reached about 58.06%, 86.56% and 99.56% at budgets 512, 2,048 and 8,192. These are quality and work observations for a flat pool with a 1,000-candidate heap. They do not establish a latency ranking between an implemented native B and a hypothetical C.

7.4 What was not validated

The study does not validate:

- real recall at a billion documents;
- a routing-recall model across arbitrary cluster counts;
- a normal tail for every query or a fixed percentage of Matryoshka prefix information;
- a concrete hash enumerator's timing, peak queue size, or SIMD speed;
- float-vector recall, human relevance, or final answer accuracy from binary recall alone.

The supplied calculator notes mixed float top-100 references at different dimensions with the binary-score reference. That mismatch cannot validate one common recall curve. The new report uses the same binary truth where it makes direct index comparisons and labels the other evidence.

7.5 How to extend the checks fairly

Freeze the representation, query set, filtering task and reference score first. Tune on one query set; check selected configurations on a different one. For every method, retain failures and empty returns, and record counts as well as times. For C, implement the actual enumerator before claiming its measured latency. For B, record split words, leaf words, documents scored and peak frontier memory separately.

At each target recall, compare measured latency quantiles using configurations that actually reach that target. Use the same query population and report the uncertainty of quality estimates. Query-level resampling is appropriate; counting repeated timings as independent quality samples is not. A few measured cases are useful evidence about mechanisms, not a complete characterization of every data distribution.

8 Bring the formulas together

The three methods share query embedding, preparation, routing, filtering, and the optional final reranker. Their difference is the work used to obtain candidates from the selected pool. This is the right boundary for comparing the index algorithms without hiding common costs.

8.1 Latency, one method at a time

Using the earlier definitions,

$$T_A = T_{\text{common}} + P c_{\text{range}} + N_{\text{elig}} G c_{\text{lut}} + H_s(N_{\text{elig}}, R) c_{\text{cheap}} + T_{\text{rank}}(R) + T_{\text{finish}}(R'), \quad (55)$$

$$T_B = T_{\text{common}} + T_{\text{order}} + T_{\text{roots}} + \sum_j s_j W_j c_{\text{split}} + \sum_j \ell_j W_j c_{\text{leafword}} \\ + V c_{\text{node}} + F G c_{\text{lut,leaf}} + H_s(F, R) c_{\text{cheap}} + T_{\text{rank}}(R) + T_{\text{finish}}(R'), \quad (56)$$

$$T_C = T_{\text{common}} + T_{\text{keyprep}} + \sum_j \left(E_j c_{\text{enum}} + C_j^{\text{copy}} c_{\text{copy}} + H_j c_{\text{lookup}} + F_j c_{\text{post}} \right) \\ + \sum_j F_j G_{C,j} c_{\text{lut,gather}} + H_s(F, R) c_{\text{cheap}} + T_{\text{rank}}(R) + T_{\text{finish}}(R'). \quad (57)$$

$G_{C,j}$ is the candidate scoring work actually needed after reusing the key contribution; it can be zero for a full key with its exact score already known.

All cluster sums in query work run over $j \in \mathcal{P}(q)$. If C uses a merged stream, add its explicitly counted stream-selection cost. Different execution choices can share preparation or reuse buffers; record those choices rather than charging duplicated work.

8.2 An index speedup is not the same as an end-to-end speedup

If two configurations have identical embedding and final-stage costs, write their shared time as T_0 . Replacing index time t_A by t_B gives speedup

$$\frac{T_0 + t_A}{T_0 + t_B},$$

not t_A/t_B . Even removing index time entirely cannot exceed $1 + t_A/T_0$ when $T_0 > 0$. If candidate counts or final-vector fetches differ, those costs are no longer shared and must be evaluated separately. This is why cached-query index timings and complete text-query timings should both be reported rather than substituted for one another.

8.3 Persistent storage

For the reference layouts selected in this report,

$$M_{A,\text{index}} = M_{\text{base}}, \quad (58)$$

$$M_{B,\text{index}} = M_{\text{base}} + db_w \sum_{j=1}^C \lceil n_j/w \rceil, \quad (59)$$

$$M_{C,\text{index}} = M_{\text{base}} + N b_{\text{ref}} + \sum_{j=1}^C \lceil U_j/\alpha \rceil [b_k(h_j) + 2b_o + b_{\text{slot}}]. \quad (60)$$

Storage sums run over *all* clusters, not only probed clusters. Increasing probes usually changes query work and scratch, not the already built index size. For literal prefix relaxation, substitute the trie directory and query-work formula from Equation (32).

A different C layout can remove the additional posting permutation; it cannot remove the information needed to map results back to documents.

8.4 Peak query RAM and the serving budget

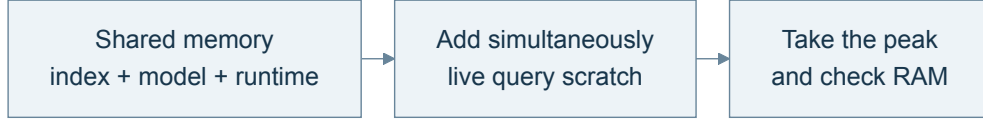


Figure 6: Persistent index data, runtime reserve and simultaneous scratch are separate terms. Sequential stage buffers need a lifetime analysis.

The search-stage scratch terms are

$$Q_{A,\text{search}} = Q_{\text{base}} + Q_{\text{scan workspace}}, \quad (61)$$

$$Q_{B,\text{search}} = Q_{\text{base}} + Q_{\text{order}} + \max_t [\gamma_B Q_{B,\text{frontier}}(t) + Q_{\text{current}}(t) + Q_{\text{children}}(t)], \quad (62)$$

$$Q_{C,\text{search}} = Q_{\text{base}} + \max_t \left[\sum_j \gamma_C Q_j(t) b_{\text{enum},j} + Q_{\text{stream merge}}(t) + Q_{\text{posting buffer}}(t) + Q_{\text{gather}}(t) \right]. \quad (63)$$

Small iterator state is included in workspace terms. Embedding and final reranking can have larger scratch than search, so use Equation (12) to obtain the full query peak. With resident index size $M_{m,\text{resident}}$ for method m , the RAM constraint is

$$M_{m,\text{resident}} + M_{\text{model}} + M_{\text{runtime}} + \max_t \sum_{u \in \mathcal{A}(t)} Q_{m,u}(t) \leq M_{\text{RAM}}. \quad (64)$$

M_{model} is the shared resident embedding-model memory if encoding runs locally; it is zero in this memory domain when an external service performs encoding. It is not multiplied by the number of queries.

For a conservative bound, replace the sum by $JQ_{m,\text{peak},\text{max}}$. A loose upper bound that exceeds RAM does not prove that a tighter implementation cannot fit. An observed peak below the limit does not prove that every possible query fits. These are different claims.

8.5 A compact comparison table

	A: full scan	B: bitplanes	C: key enumeration
Additional index	none beyond common layout	$db_w \sum_j W_j$	postings and key directory
Principal work	every eligible code	split masks, leaf masks, full scores, frontier	empty and occupied keys, enumeration, postings, full scores
Variable counts	N_{elig}	s_j, ℓ_j, f_j, V	$H_j, F_j, E_j, Q_j^{\text{max}}$
Largest scratch concern	selection or batch score buffers	many full-width pending masks	many pending flip sets or candidate gather buffers
Local exactness	scan all eligible rows	valid bounds, sufficient traversal, correct ties	full-key weighted order or a valid full-score bound

	A: full scan	B: bitplanes	C: key enumeration
Approximation control	routing probes; candidate/rerank boundary	node budget, visit policy, earlier routing	prefix width, candidate or key budget, earlier routing
What must be measured	kernel cost, routing quality	skipped scores versus bitmap/queue cost and recall	empty-key volume versus useful candidates, timing and recall

8.6 The variables most likely to move the tradeoff

Documents per physical cluster, n_j . A grows roughly linearly with eligible row count. In B, a larger cluster also widens every active bitmap. In C, more documents per key can reduce the number of empty probes needed, while increasing posting and scoring work.

Binary dimension, d . Longer codes increase score terms and stored bits. B may gain more from avoiding full scores, but also stores more planes and can traverse more coordinates. For full-key C, the number of possible keys grows exponentially. For short-key C, h and d are separate parameters.

Probes, P . More clusters can recover missed neighbors. They also increase eligible rows, root masks, active key streams and candidate competition. Cluster sizes and the query’s routing behavior matter, not just the number P .

Leaf size and node budget. A small leaf threshold can trade scoring for more mask work. A finite node budget caps visits but does not specify how many useful leaves are reached. The best choice depends on the query and data, not a general monotonic recall formula.

Lookup width and candidate target. A short key has denser postings and a smaller key space, but leaves more score information out. A longer key can improve candidate discrimination and simultaneously make empty-key enumeration costly. A strong prefix can improve recall; that does not remove the enumeration cost.

Kernel and memory system. SIMD, cache residency, batch size, alignment, layout and thread count change the effective coefficients. Keep them fixed for an algorithm comparison, or report their changes as a separate experiment.

8.7 Choosing configurations at fixed N and RAM

Let θ_m contain the chosen method’s tunable parameters and let r_0 be a recall target. Choose the latency objective explicitly, for example median or p95 across queries. A valid optimization statement is

$$\theta_m^* \in \arg \min_{\theta_m} \text{Quantile}_p[T_m(q; \theta_m)] \quad (65)$$

subject to the RAM condition in Equation (64) and

$$\mathbb{E}_q[\text{Recall}@K(q; \theta_m)] \geq r_0. \quad (66)$$

The expectation specifies an average-recall target. A requirement that almost every query reach r_0 is stronger and needs a different probability constraint.

Listing 12: A defensible finite-grid experiment.

```
ChooseConfiguration(corpus, queries, fixed_reference, RAM_limit, targets):  
    build or cache every configuration's required index  
    compute exact reference results once for each query  
    on tuning queries:  
        for every configuration in the declared grid:  
            measure latency, quality, work counts and peak RAM  
            keep failures and empty returns in the record  
    choose qualifying configurations for each target  
    on separate test queries:  
        evaluate those choices without retuning  
    report lowest tested latency only where the target and RAM checks pass  
    distinguish measured results from untested model predictions
```

A mathematical model can reduce the grid and explain its results. It cannot manufacture the unknown query/document distribution or the true hardware coefficients. A “best estimated point” from a model is a proposed experiment, not a verified engineering choice.

If d is also varied, keep the quality reference fixed across choices, for example the same long-vector top K or the same relevance labels. Recomputing a different binary ground truth for every d makes a tiny representation look perfectly accurate against its own changed task.

Final interpretation

A pays to score the selected pool. B pays to discover which scores can be avoided, and dense bitmaps can make that discovery expensive. C pays to discover which keys contain useful candidates, including empty keys and queue work. Each can be exact under a suitable stopping rule. Which is fastest at a recall target is determined by the data, the representation, the visited work, the memory layout and the hardware—not by one operation count or one assumed normal distribution.

A A finite-sample view of prefix recall

The population-tail approximation is not the only mathematical definition available. This appendix shows what an exact finite- N_* calculation would require, under independent, identically distributed document score pairs and no score ties. It is useful for understanding why knowing only a correlation is insufficient.

For a particular document with full score s and prefix score t , classify another document into four events:

$$\begin{aligned} p_{11}(s, t) &= \Pr(S > s, T > t), & p_{10}(s, t) &= \Pr(S > s, T \leq t), \\ p_{01}(s, t) &= \Pr(S \leq s, T > t), & p_{00}(s, t) &= \Pr(S \leq s, T \leq t). \end{aligned}$$

Among the other $N_* - 1$ documents, let (a, b, c, e) count these categories. A multinomial distribution counts how many independent documents land in each of the four categories, using those probabilities. The chosen document belongs to the full top K when $a + b \leq K - 1$, and belongs to the prefix top R when $a + c \leq R - 1$. Therefore, with $e = N_* - 1 - a - b - c$,

$$P_{K,R}(s, t) = \sum_{\substack{a, b, c \geq 0; \\ a+b \leq K-1; \\ a+c \leq R-1 \\ a+b+c \leq N_*-1}} \frac{(N_* - 1)!}{a!b!c!e!} p_{11}^a p_{10}^b p_{01}^c p_{00}^e. \quad (67)$$

If $F_{S,T}$ is the joint distribution of one document's scores, every document has the same event probability under the identical-sampling assumption. Summing over all N_* documents and dividing by K gives

$$\mathbb{E}[\text{Recall@ } K] = \frac{N_*}{K} \int P_{K,R}(s, t) dF_{S,T}(s, t). \quad (68)$$

This assumes the candidate stage returns exactly the prefix top R and the final stage uses the same full score. Retrieving whole tied prefix keys changes that candidate rule; ties can be handled by augmenting the ordering with document IDs or by explicitly integrating their selection probabilities.

Equation (68) is much harder to evaluate than the conditional-normal approximation. More importantly, it still needs $F_{S,T}$. Matryoshka training, a code length, a cluster count and a RAM budget do not uniquely specify that distribution. Exact mathematics can organize unknown inputs; it cannot infer them from the algorithm's name.

B Checklist for an honest numerical implementation

1. Fix the score and tie rule. Keep original-float, truncated-float, binary-score and human relevance references distinct.
2. Check the score identity on small vectors and enumerate the actual top K .
3. Check that the weighted-subset stream emits every subset once, in nondecreasing penalty, and processes the final boundary tie before claiming exactness.
4. Count split words and leaf words separately. Use each cluster's physical width, including padding; do not substitute the number of active documents for the bitmap length.
5. Record the ownership of codes, external IDs, posting references and key storage. Any shared or reordered array must be stated before removing its bytes from a formula.

6. Count temporary state at its lifetime peak. A queue-capacity bound and an observed allocation peak are different quantities.
7. Keep key attempts separate from occupied keys and returned postings. Expected first hit is not the same threshold as finding R candidates.
8. Price centroid float arithmetic separately from binary score lookups. Charge selection, allocation, copying and gathers where the implementation performs them.
9. Use either a calibrated combined cost or a disaggregated operation model for each block. Do not count the same memory and selection work in both.
10. Test statistical predictions on queries not used to estimate their parameters. Evaluate recall after the actual candidate and tie rules, not merely a probability of finding some key.

C Floating-point arithmetic and exactness

The score and bound proofs use real arithmetic. A numerical implementation must ensure its computed upper bounds remain conservative under rounding. The local C++ reference includes a rounding allowance for bound pruning and a deterministic row-ID tie order. A report should not claim exactness merely because an unguarded floating-point comparison resembles the algebra.

This does not require inventing a collection of hypothetical production patches. It requires checking the numerical path used by the implementation and testing it against a full reference scan on the same inputs. The small examples and archived candidate checks provide such evidence for their stated scopes; they do not prove every future implementation correct.

D Optional expansion of the embedding cost

The main formulas keep T_{emb} and Q_{emb} as measured model-specific functions. If a more detailed encoder estimate is needed, its architecture must be another set of inputs, not an unexplained constant.

For an illustrative standard dense Transformer encoder, let n_t be query tokens, L_e the number of layers, m hidden width, f feed-forward width, and H_a attention heads. For one layer with ordinary two-matrix feed-forward computation, multiply-add contributions are

$$U_{\text{layer}} = 4n_tm^2 + 2n_t^2m + 2n_tmf. \quad (69)$$

The first term comes from the query, key, value and attention-output projections. The second comes from attention-score and attention-value matrix products. The third comes from the feed-forward expansion and contraction. Activation, normalization and softmax operations add separate elementwise terms. A different block, gated feed-forward network or sparse attention requires different counts.

A corresponding parameterized time model is

$$T_{\text{emb}} = T_{\text{tokenize}} + L_e(U_{\text{layer}}c_{\text{enc,mac}} + U_{\text{elementwise}}c_{\text{enc,elt}}) + T_{\text{pool/project}}.$$

These effective coefficients depend on batching, device kernels, precision and memory behavior. Alternatively use the resource bound in Equation (19), without adding it on top of coefficients that already include the same work.

A non-fused attention implementation may materialize $H_a n_t^2$ attention values. With bytes per activation b_a , an explicit scratch model can be written

$$Q_{\text{emb}} = b_a(\eta_m n_t m + \eta_f n_t f + \eta_a H_a n_t^2) + Q_{\text{kernel}},$$

where the η coefficients count simultaneously live buffers, not time. Fused attention can avoid materializing the square attention matrix, so its chosen buffer counts differ. Layer workspaces can be reused during inference; multiplying a per-layer peak by L_e is not a general inference-RAM formula. Resident model weights are shared and counted separately.

A Matryoshka prefix normally reuses an already computed embedding. Halving the search prefix d does not automatically halve the encoder’s layer count or hidden width. It reduces the subsequent representation and search work unless the actual encoder implementation also changes.

E Reproduction and parameter ledger

The editable source is `report.tex`, with chapter files in `sections/`. Mermaid sources and rendered diagrams are in `figures/`. The build instructions are in the adjacent `README.md`. The arithmetic checks are implemented in `verify_math.py`. They validate the worked examples, exact subset ordering, binomial conditional-recall example, word widths, selected storage identities, and quoted measured rows. The statistical evidence is preserved with its source hash.

The following ledger identifies the inputs a future calculator must obtain rather than infer from one document count.

Input family	Parameters or counts	How to obtain them
Representation	D, d, d_r, d_f , preparation and score rule	fixed experimental definition; embedding/model metadata
Retrieval target	K, R , filter, reference score, tie rule	fixed task definition
Cluster layout	$C, n_j, \mathcal{P}(q), e_j$	built index and per-query routing/filter traces
Storage widths	$w, b_w, b_{\text{id}}, b_o, b_c, b_f, b_s$	actual types, packing and alignment
Baseline scoring	$g, G, H_s, c_{\text{lut}}$	source code, work counters and kernel measurements
Bitplane search	$L, B_{\text{node}}, s_j, \ell_j, f_j, V$	configuration plus query traces; not budget alone
Bitplane RAM	frontier masks, current/child buffers, capacities	allocation/capacity measurements or an explicit bound
Key index	$I_h, h_j, U_j, \alpha, b_k, b_{\text{ref}}$	chosen lookup coordinates, index build and slot layout
Key search	$H_j, F_j, E_j, Q_j^{\text{max}}$, stopping rule	actual key enumeration, posting and frontier traces
Prefix-trie variant	$Z_j, b_{\text{trie}}, u_j$, posting ranges	actual stored nodes, path and range visits
Hardware	compute rates, bandwidth, cache regime, SIMD, threads	calibrated for the actual workload and machine
Optional final stage	fetch policy, R', u_f, d_f , rerank score	execution pipeline and storage-residency decision

Input family	Parameters or counts	How to obtain them
Statistical model	query/data distribution, covariance, score tails	declared synthetic assumptions or measured and checked estimates
Serving RAM	$M_{\text{resident}}, M_{\text{model}}, M_{\text{runtime}}, J, Q_{\text{peak}}$	ownership/lifetime analysis and process measurements

A numerical example can assign values to these parameters. The equations themselves remain parameterized, so changing the workload changes the inputs instead of requiring a different explanation of the algorithm.

References

- [1] Exa Team. *How we built a web-scale vector database*. 17 December 2024. <https://exa.ai/blog/building-web-scale-vector-db>.
- [2] A. Kusupati et al. *Matryoshka Representation Learning*. 2022, revised 2024. <https://arxiv.org/abs/2205.13147>.
- [3] M. Norouzi, A. Punjani, D. J. Fleet. *Fast Exact Search in Hamming Space with Multi-Index Hashing*. 2013, revised 2014. <https://arxiv.org/abs/1307.2982>. This concerns Hamming search; it is not a weighted-search guarantee for the methods here.
- [4] Local project evidence. `cpp/index.cpp`, `cpp/index.hpp`, `results/scaling-msmarco-1m/`, `results/fiqa-2026-09-09-dense/`, and `calculator/recall-validation.json`. Reproduction and source hashes are described in Section 7.